

Harsh Desai

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EDUCATION

University of Illinois Chicago

B.S. Computer Science

Expected May 2027

TECHNICAL SKILLS

Languages: Python, Java, C++, C, JavaScript, HTML/CSS, Swift, SQL, Lua, C#, Qiskit

Frameworks & Tools: React, Next.js, Spring Boot, MongoDB, Git/GitHub, Jupyter Notebook, Qbraid, Xcode, Unity, IntelliJ, VS Code

EXPERIENCE

Quantum Algorithms Researcher | University of Illinois Chicago

Aug 2025 – Mar 2026

- Reduced circuit complexity by investigating Jordan-Wigner vs. Bravyi-Kitaev fermion-to-qubit mappings, benchmarking simulation efficiency across multi-qubit systems while maintaining equivalent time-control accuracy.
- Designed experimental frameworks in Qiskit to benchmark qubit mapping approaches, producing reproducible comparisons across multi-qubit quantum circuit configurations.

Quantum Software Development Intern | University of Illinois Chicago

Summer 2025

- Doubled measurement accuracy across variable configurations by building Python scripts to compute expectation values for Quantum Harmonic Oscillators and multi-qubit systems.
- Produced reproducible experimental results analyzing behavioral shifts in Quantum Harmonic Oscillators under both incremental and drastic variable changes.

Computer Science Instructor | Code Ninjas, Long Grove

2022 – 2024

- Developed and delivered curriculum for 8+ students aged 8–13 covering programming fundamentals, game design, and AI concepts using Roblox Studio, Scratch, and Code Combat.
- Designed custom lesson plans and slide decks for rotating class cohorts, adapting material for varying skill levels.

PROJECTS

CurtainCall · Next.js · React · Spring Boot · MongoDB · Tailwind CSS

Feb 2026

- Built a Tinder-style artist networking platform with a swipe-based matching system, designing 12+ RESTful API endpoints with Spring Boot to support scalable user connections, endorsements, and discovery.
- Architected MongoDB collections around user relationships and endorsement logic, structuring schemas to support 1,000+ concurrent users while eliminating redundant database calls. Live: [CurtainCall Live Demo](#)

SnapLink · React · Spring Boot · MongoDB · Google Maps API

Feb 2025

- Built a geo-tagged task discovery platform connecting small businesses with local job-seekers in real time, integrating the Google Maps API with dynamic geolocation filtering for hyper-local opportunity matching.
- Designed a full-stack architecture with RESTful Spring Boot APIs and MongoDB, enabling scalable task posting, status tracking, and location-based querying. Live: [SnapLink Live Demo](#)

CountingCards · HTML · CSS · JavaScript · Bootstrap

Feb 2026

- Built a gamified card-based study system with dynamic card cycling, interactive session tracking, and a responsive Bootstrap UI—designed to replace passive reading with active recall loops. Live: [CountingCards Public Link](#)

Lucky D's Gaming Cafe Website · React

Jun 2022

- Independently designed and deployed a production-ready React site for a local gaming café, enabling customers to explore events and services before visiting. Live: [Website Public Link](#)

AWARDS & LEADERSHIP

- IBM Quantum Badge of Excellence** – Completed all labs in the Qiskit Global Summer School 2025 with 100% accuracy.
- Freshman Mentor Program** – Selected as 1 of 100 mentors from 2,000+ applicants; recognized as a top mentor by program leadership.
- IEEE Executive Board** – Web development board member; leading redesign and modernization of the IEEE UIC chapter website.